

## V1: Base:

- Inner dimensions are  $128.8 \times 86.5 \times 18$  mm to support a Tbird Multi-well plate which has dimensions  $127.8 \times 85.5 \times 16.6$  mm with the lid on
- There is a small ledge on the bottom that the plate sits on.
- Groove around the top of the base, <sup>depth = 1.0 mm</sup> this is designed to fit a 4mm Gasket  
(based on a little bit of searching, Silicone Gaskets are used inside autoclaves since they can withstand high temperatures.)
- 4 M4 bolts used to clamp lid to base
- Bottom has a 1.1 mm recess with a ledge that allows for the insert of a Glass Petri which can be sealed around the edges using a suitable seal (saw somewhere recommended use silicone but will check)
- Hex shaped recesses to allow Hex Nut for M4 bolt to slot in easily

## LID:

- Holes to support M4 bolts
- Lid has a 1.1 mm recess with ledge that allows the insert for a plate (Solid / glass)
- Bottom has 4mm <sup>→ check original sizes</sup> groove with depth 1.0 mm

that matches base groove for the Gasket

total gasket groove height =  $2.8 \text{ mm}$   $\rightarrow$  Allows for Gasket to be squished for seal

Main Points: Very basic Version with just sealing properties.

- No provision for temp control (ASK how i should Include in model, if heating elements are inside)
- No Provision for future Gas control
- No Provision for dish compatibility
- No Provision for microscope mount
- Secured by bolts, so may be not user friendly (would a latch mechanism work?)  
(what tools do you have)
- Is there a BSL-3 containment document you use to check containment specs

## Design changes

- Glass pane on Inside Not outside

- Glass on wells used as sealing pane?

Instead of having a sealed chamber with a glass pane, use the plate glass as the sealing unit

- Overcenter mechanism!

want to be able to flip the chamber upside down but keep dish upright to allow for top-down M-scrape

- mating seal design (top)

Perfect touch



Sealing groove  
(crush seal)

Inside can be not perfect touch

This eliminates variance in sizes

- possibly, make disposable by permanently attaching bump to plate, then don't worry about getting seal

- oversize Bolt holes for clearance